35. THE NEURAL PLATE

DURATION : 21:32 STUDY SHEET

COURSE SUMMARY

This video delves into the fascinating dynamics of the neural plate and the notochordal process, essential for embryonic formation. By studying how the notochord influences the curvature of the neural plate and the formation of the neural groove, students will discover the fluidic and vascular interactions that shape the development of the neural tube. The importance of elements such as mesenchyme and molecular compounds like Sonic Hedge-Hog (SHH) and Bone Morphogenetic Proteins (BMP) will be discussed, highlighting their role in dorsalisation and cellular induction. In practice, osteopaths will learn to use this knowledge to rebalance the nervous system and release tensions, while also measuring the impact of the environment on embryonic development.

THE NEURAL PLATE

INTRODUCTION TO THE NOTOCHORDAL PROCESS

When examining the **sacrum**, one observes the **notochordal process**, a descending wave that acts as a fulcrum. This notochordal process directly induces the impulse on the **neural plate**, orienting and organising the force applied to it. In just three days, the growth of the embryo manifests, with a descent of **Hensen's node** relative to overall growth.

FORMATION OF THE NEURAL GROOVE AND VASCULAR SYSTEM

The beginning of the **neural groove** appears, accompanied by the **pharyngeal** and **cloacal** membranes. The notochordal process plays a crucial role, allowing the neural plate to curve and form a groove. This process occurs synchronously with a lateral and fluid reorganisation.

At the time of neural groove formation, a fluid organisation occurs, marked by the appearance of **fine aortic capillaries**. For this lateral fluid process to be effective, several components must be present:

- A trajectory
- Particles

- Vacuolations
- Metabolic concentrations

These elements, present in the **mesenchyme**, create a lateral vascular trajectory. Thus, the tube gradually closes while establishing a vascular trajectory in the **mesoderm**.

INTEGRATION OF AMNIOTIC FLUID AND NEURAL TUBE CLOSURE

Simultaneously, the **amniotic cavity** above contains fluid. As the neural plate begins to form a tube and a groove, the amniotic fluid is integrated into the neural tube. This process is accompanied by the integration of the **external coelom** into the **internal coelom**.

When the neural tube closes completely, with the closure of the **anterior and posterior neuropores**, the integration of the coelom is total. This gives rise to a visceral box, an abdominal cavity, and a cerebral cavity, all formed simultaneously. This moment, on the **28th day**, is marked by remarkable synchronicity.

SYNCHRONICITY OF EVENTS AND THE ROLE OF THE NOTOCHORD

This moment of closure is crucial. With the integration of the coelom and the aortas, tissue forms around the notochord, in metabolic retention fields. This closure is constant, and for osteopaths, it is interesting to observe these phenomena occurring at a distance but at the same time.

The precursor of this groove is the notochord. Its closure leads to basic molecular levels that slow down growth. The **Sonic Hedge-Hog (SHH)** phenomenon intervenes, influencing the dorsalisation of the system, accompanied by **bone morphogenetic proteins (BMP)**.

MOLECULAR PHENOMENA AND GENETIC INFLUENCES

Neural Crest Cells also play a role, manifesting morphological, genetic, and molecular dynamics. The observed movements, described by Deschmidt and others, reveal phenomena of inhibition or induction on molecular bases of a genetic type.

The notochord influences the cells associated with it, causing an induction phenomenon. This information will be crucial for neurulation, influencing the **ventral midline** and causing the **dorsal midline**. The growth of the dorsal midline re-influences the anterior midline, thus creating a ventral, dorsal, and anterior **midline**.

MOVEMENT, ENVIRONMENT, AND THERAPY

Morphogenetic poles, such as types 4 and 7, participate in dorsalisation. These molecular phenomena are supported by scientific bases, explaining the organised programme at this level. Movement reorients development, highlighting that the environment, rather than genes, is the driving force of this process.

From a practical perspective, it is essential to understand the environment in which the patient evolves. The notochordal axis serves as a reference for releasing the fluid side of the brain, hydrating, and eliminating energetically dried-up areas.

AMNIOTIC FLUID AND ITS SIGNIFICANCE

We will focus on the first moment of neural tube closure. This involves a rebalancing of the nervous system, both sympathetic and parasympathetic. It is crucial to remember three elements: the **ectodermal** tissue and its epithelial cells, the movement induced by the notochord, and the lateral growth that forms the groove.

We begin to distinguish the cells that will form the neural tube, encompassing the spinal canal and the brain, while laterally, other cells will form the epidermal epithelial tissue.

NEURAL CREST CELLS

Another important structure appears: the anlage of the **neural crest** tissue. Neural crest cells, in red, blue, and black, not only contribute to neural tube closure but also form the peripheral nervous system. They interact with the mesoderm to create ecto-mesodermal tissues, such as the face and skull bones.

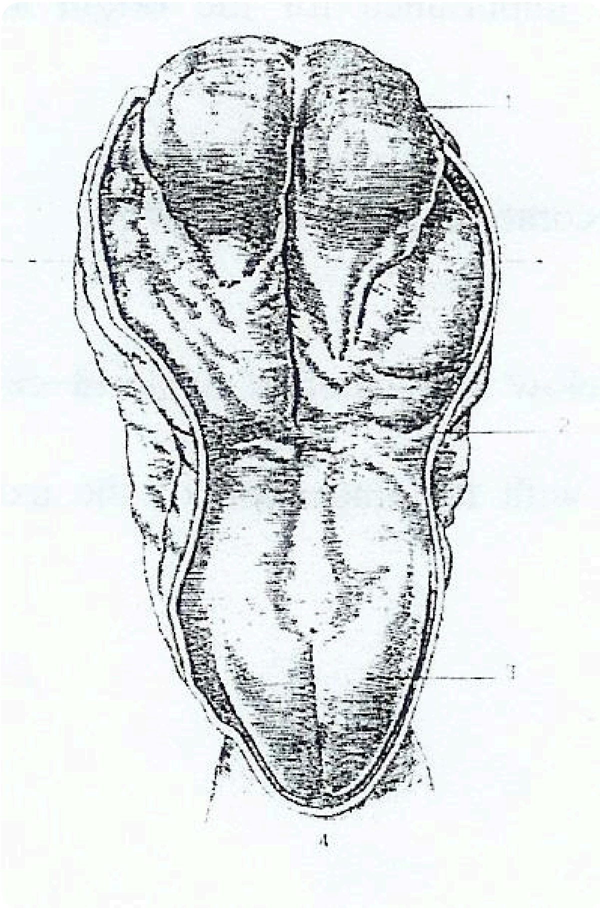

FIG. — Embryonic neural crest

The migrating neural crest draws in the rhombencephalic neural crests to form the facial system. The cephalic brain divides into **prosencephalon**, **mesencephalon**, and **rhombencephalon**, each creating a neural crest.

ROLE OF INFORMATION AND MOVEMENT

This tissue is informed and transmits information, connecting the belly to the periphery and vice versa. Sensors, such as those for atmospheric pressure and light, provide essential information. The ectoderm, upon receiving information from the notochord, does not grow randomly.

The tissue cells differentiate into three types:

- Those forming the neural tube.
- Those forming the neural crests.
- Those forming the epidermal epiblast.

These tissues are fundamental for the development of the peripheral nervous system, mesenteric plexuses, and other bodily structures.

LINKS BETWEEN AMNIOTIC FLUID AND DEVELOPMENT

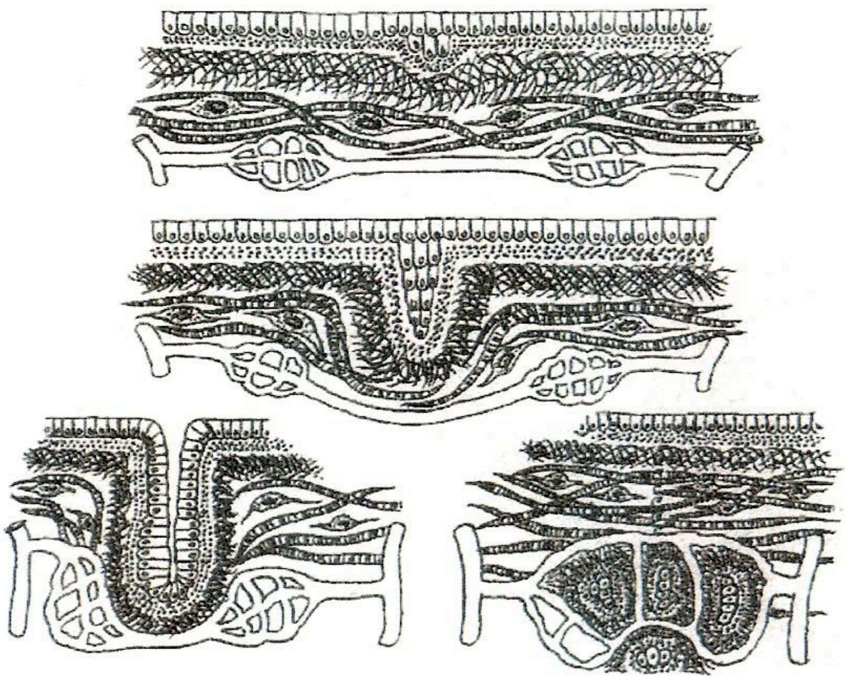

FIG. — Intestinal villi development

Amniotic fluid, rich in **building proteins**, plays a crucial role. This fluid, present inside and outside, is linked to the quality of the skin and cerebrospinal fluid. The latter is essential for body movement and cleansing.

Embryonic development involves several cavities, including the **blastocoel** and the **amniotic cavity**. The child ingests amniotic fluid, integrating information about its development. After the 28th day, once the neural tube is closed, the child begins to eliminate its first pro-urine, creating a link with the aortic system.

NEURAL TUBE CLOSURE: ANATOMY AND IMPLICATIONS

Sonic Hedgehog (SHH) is crucial at the genetic level, as are **Cell Adhesion Molecules (CAM)**, which play a role in cell adhesion. Neural tube closure occurs at a time when the cephalic part is more developed than the caudal part.

This closure is decisive for the development of the embryo, connecting the prosencephalon, mesencephalon, and rhombencephalon. The notochord, below, influences the development of the nervous system.

THERAPEUTIC TOOL: REBALANCING THE SYSTEM

We will integrate all metabolic fields for the ectomesenchyme, taking into account morphological **building blocks** and growth factors. By visualising the closing movement of the neural tube, we can use this first encounter between the left and right sides as a therapeutic tool to rebalance the system.

Once the anterior neuropore is closed, it becomes the **lamina terminalis**, marking a significant impulse in bodily development.